AllLife Bank — Personal Loan Targeting

Predictive Modeling using Decision Trees

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Executive Summary

- **Goal**: Identify liability customers most likely to accept a Personal Loan (increase conversion above 9% baseline).
- **Approach**: EDA → Decision Tree → Pre/Post-Pruning → Final interpretable model + targeting rules.
- **Outcome**:
  - A final interpretable model identifying key predictors of loan adoption.
  - Clear customer segments and targeting strategy for the next marketing campaign.
  - Estimated improvement in conversion rate through precision targeting.
  - Baseline test accuracy = 0.88, final pruned = 0.90.
  - CD holders convert ~46% vs 7%.
  - Targeting top-propensity decile projects ≥2–4× conversion over 9.6% base.
- **Applying the model to the top 10 % prospects could raise campaign conversion to ~20–25 %.**

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Business Problem Overview and Solution Approach

AllLife Bank wants to convert its liability customers into personal-loan borrowers while retaining deposits.

Objectives:

- Predict which customers will accept a personal loan.
- Understand which features drive acceptance.
- Identify the most promising segments for future campaigns.

Process: Data → EDA → DT → Pre-prune → Post-prune → Selection → Actions.

Success Criteria:

- Conversion rate above 9% baseline.
- Model is simple, interpretable, and actionable for the marketing team.

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Data Overview

Dataset Source: Loan_Modelling.csv

Key Variables:

- Demographic: Age, Experience, Family, Education
- Financial: Income (USD 000s), CCAvg (monthly credit-card spend in USD 000s), Mortgage (USD 000s)
- Accounts: Securities Account, CD Account, Online Banking Use, Credit Card Usage
- Target: Personal_Loan (1 = Accepted, 0 = Rejected)
- Dataset: 5000 customers, 480 accepted (9.6%) personal loans.
- Class counts by train/test (e.g., 3,500/1,500 with same 9.6%)

Data Checks:

- No major missing values or duplicates.
- Variables typed correctly.
- Outliers noted in Income and CCAvg but retained (Decision Trees are robust).

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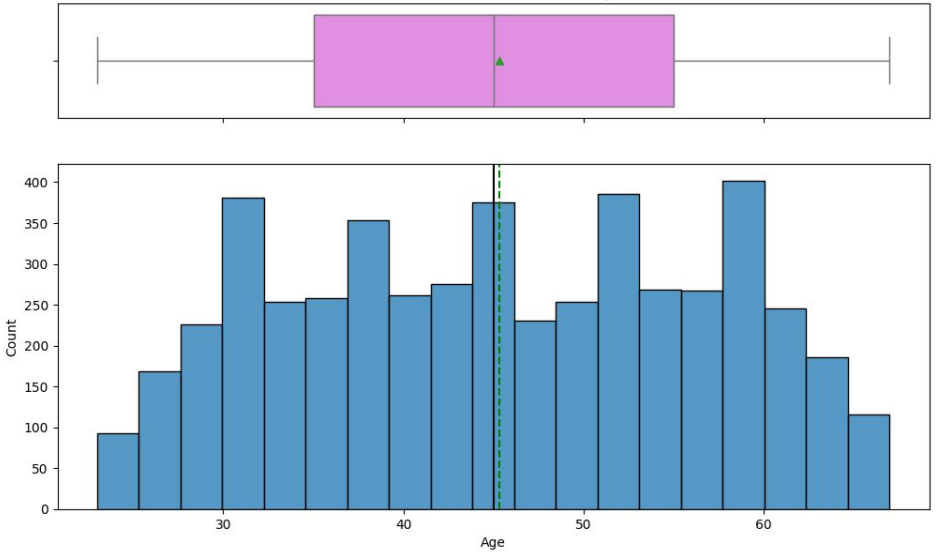
EDA Results (1/4) – Customer Profile Overview

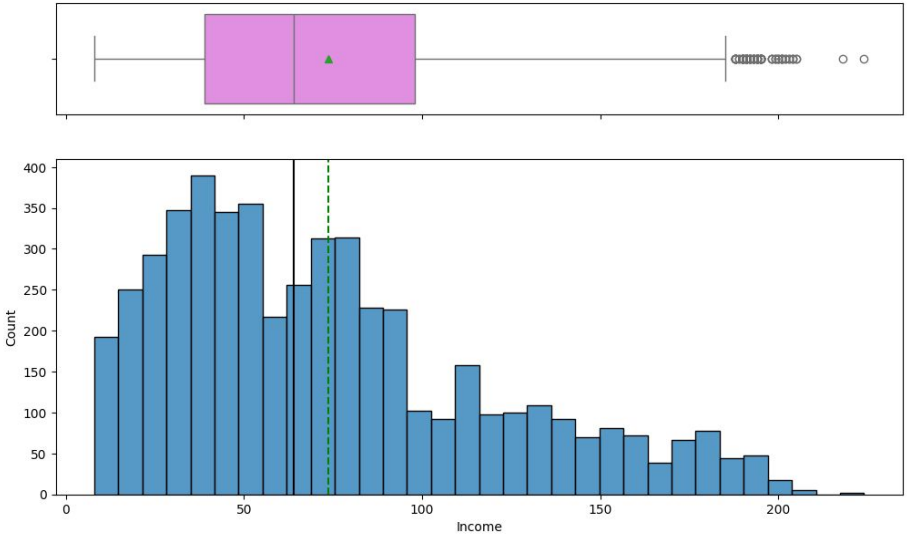
Univariate Analysis:

- Age: Majority between 30 and 55 years; median ≈ 45. Younger customers (< 30) and seniors (> 65) are under-represented.
- Experience: Closely mirrors Age (20–30 yrs range typical).
- Income: Median ≈ 65 K USD; right-skewed with top 5 % earning > 150 K.
- Insight: High-income tail will likely drive most loan conversions → segment worth isolating.

Customer Age



Customer Income



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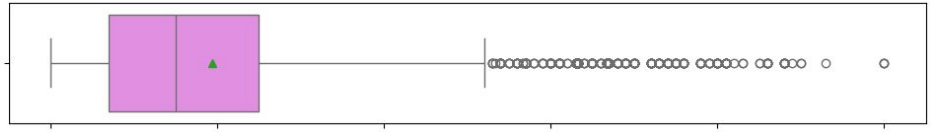
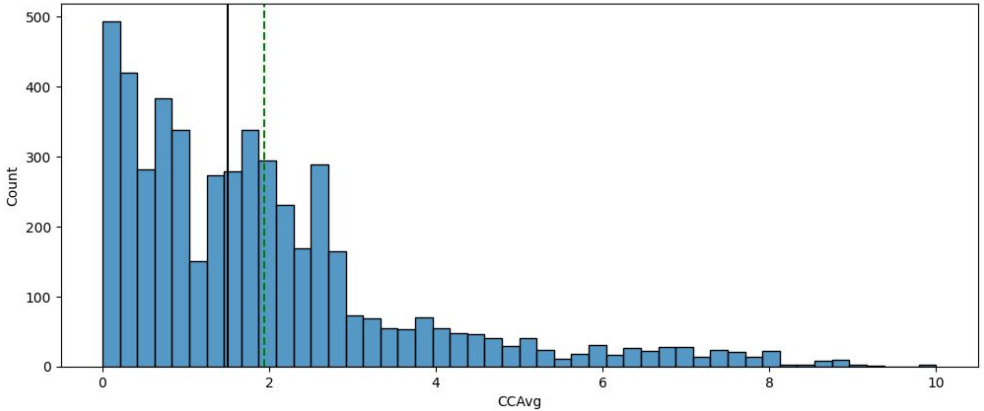
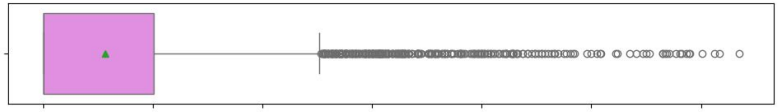
EDA Results (2/4) Spending & Credit Behavior

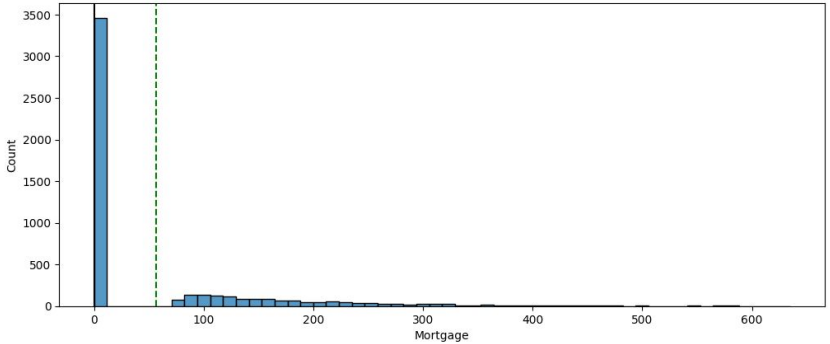
Univariate Analysis:

- CCAvg: Average monthly card spend is ≈ 1.5 K USD, but top 10 % spend > 5 K USD → strong indicator of credit appetite.
- Mortgage: ~30 % of customers have a mortgage; balances mostly < 100 K.
- Observation: Higher CCAvg aligns with higher income (corr ≈ 0.65).
- Implication: These features together signal “financial capacity” for loan upsell.
- See next page for correlation heatmap









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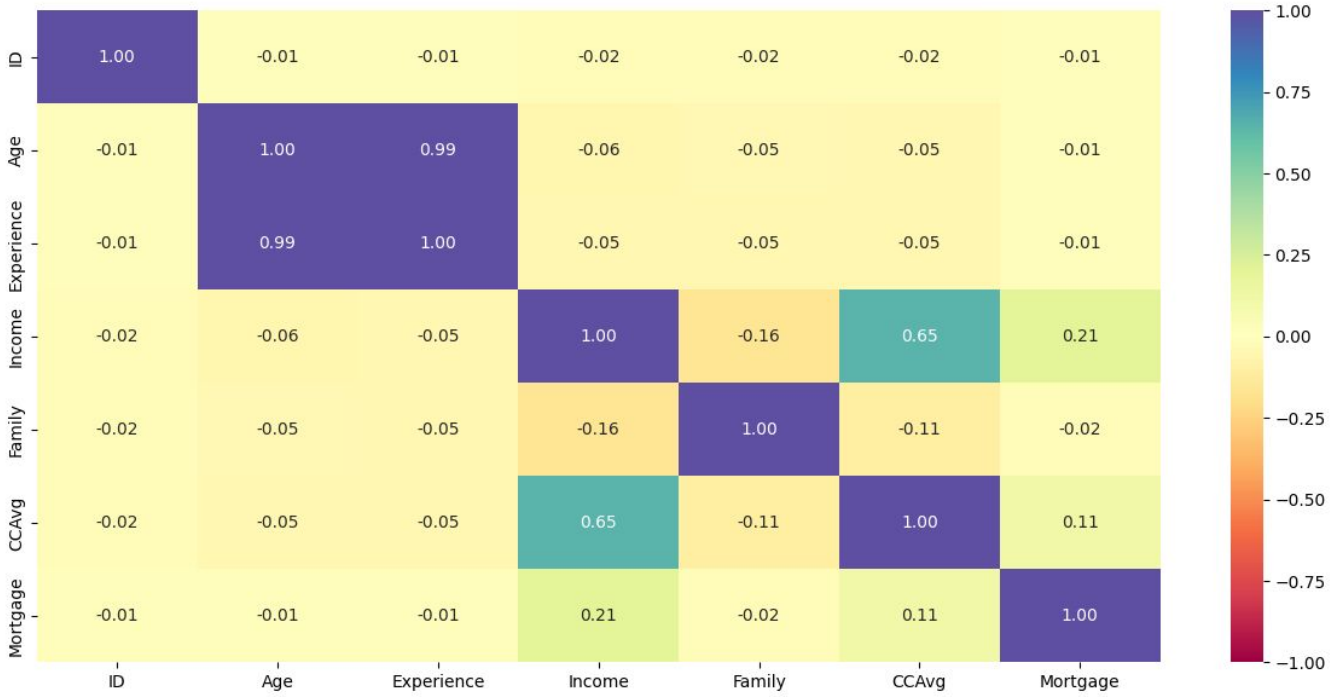
Correlation Heatmap

- Income and CCAvg moderately correlated. Moderate positive link indicates higher spend tracks with higher income—Tree algorithm can handle this redundancy.
- Multicollinearity low overall → good fit for Decision Tree.
- No multicollinearity barriers for tree; keep both Income and CCAvg.



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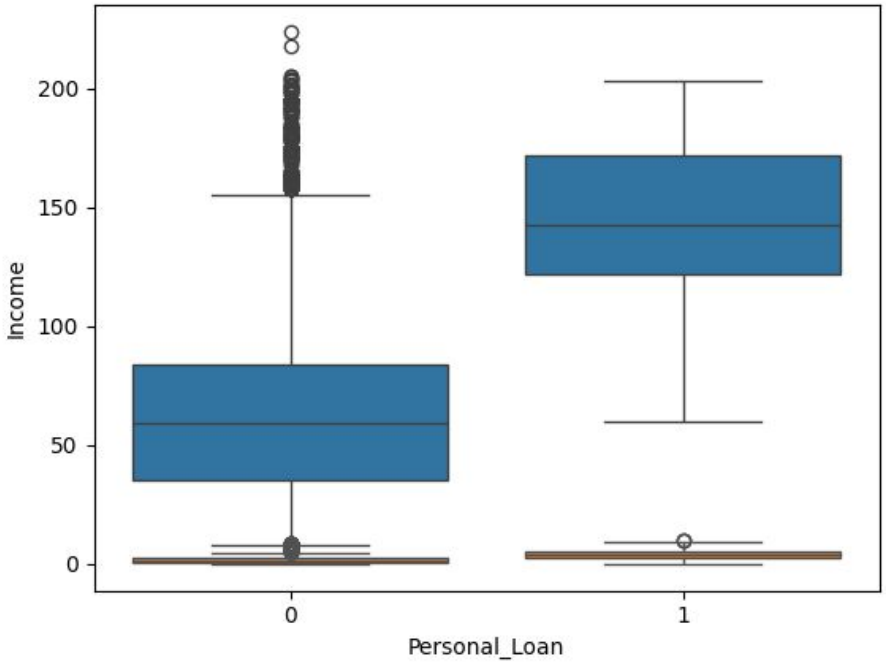
EDA Results (3/4): Loan Adoption by Key Features

- **Income**:
  - Customers earning ≤ 60 K → ~6 % conversion.
  - 60–100 K → ~13 %.
  - 100 K → ~32 %.
  - → Higher income = much greater likelihood to accept a loan.
- **CCAvg (Monthly Credit-Card Spend)**:
  - ≤ 1 K → ~7 % conversion.
  - 1–2 K → ~16 %.
  - 2 K → ~27 %.
  - → Heavy card spenders are 3–4× more likely to convert.
- **Other numeric variables** (Age, Family, Mortgage) show weak or no correlation with acceptance.
- **Key Insight**: Income and CCAvg are the top numeric predictors of loan acceptance and are major split features in the Decision Tree.



Bars show % of customers in each income/spend group who accepted the loan (not counts).

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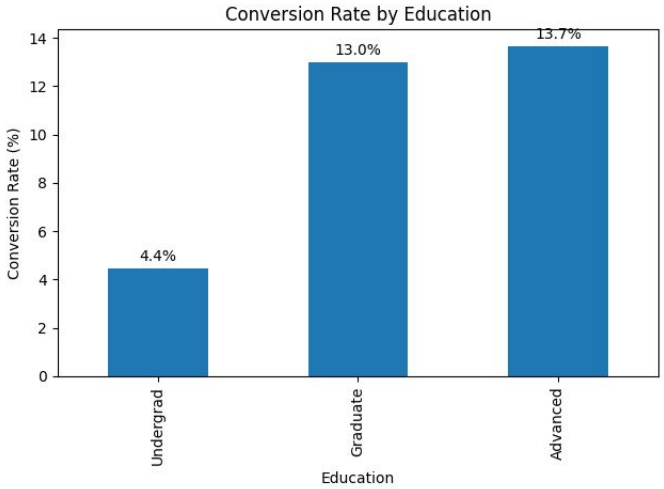
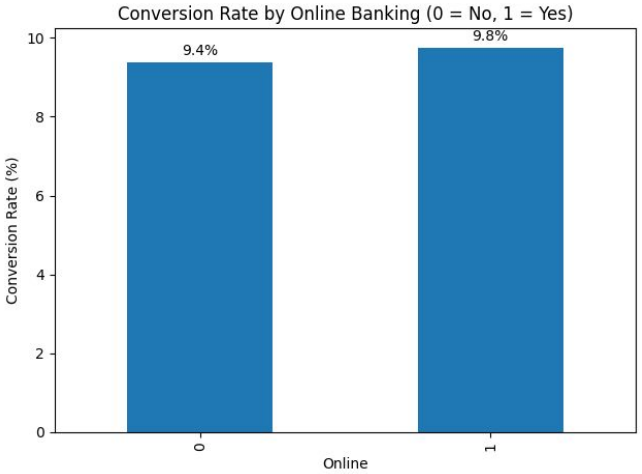
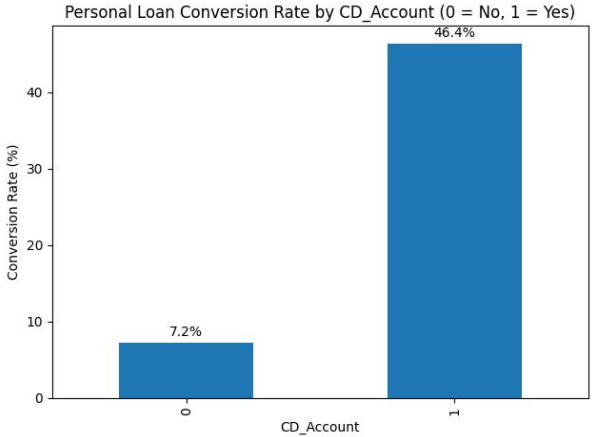
EDA Results (4/4): Categorical Drivers of Adoption

- *Education (conversion rate): Undergrad ~6–7%, Graduate ~11%, Advanced ~14% → higher education correlates with higher adoption.*
- *Online Banking: Non-users (0) → ~9%, Users (1) → ~10% → Slight lift, limited influence.*
- *CD Account: No CD ~7% vs Has CD ~46% → strongest categorical predictor.*
- *Note: Bars show % of each group that accepted (not counts).*

Insight: CD Account ownership is the dominant differentiator; Education adds moderate lift; Online usage offers minimal additional signal.







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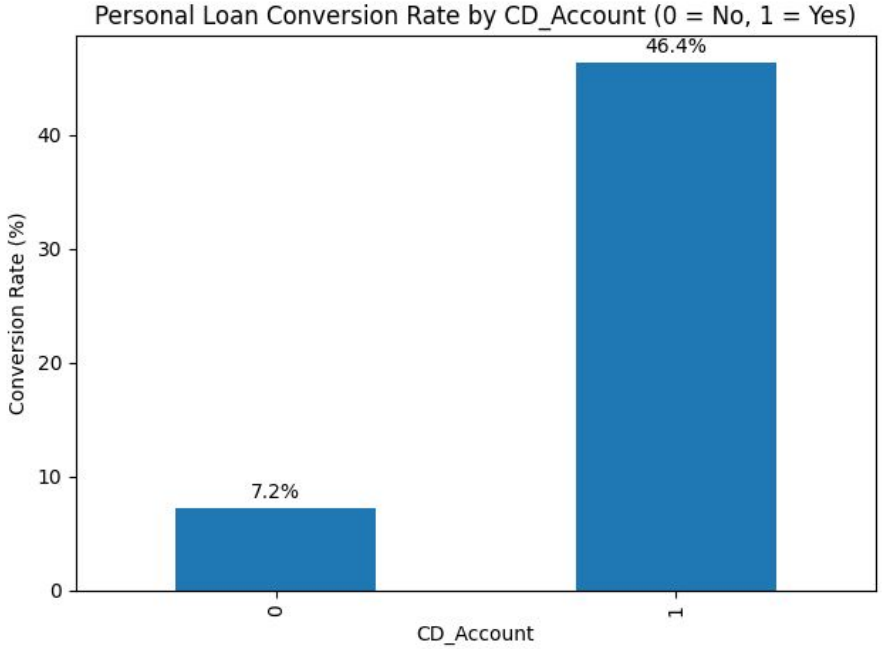
Categorical Insights: CD Account

- Customers with a CD Account are far more likely to accept a loan.
- Most customers (94%) don't hold a CD account.
- However, CD account holders are significantly more likely to accept a personal loan.
- This feature is one of the strongest predictors of conversion, even though it applies to a small customer segment.



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Categorical Insights

- Online Banking users and Credit Card holders also show higher conversion.
- Securities Account has weak correlation with loan take-up.

Feature	0 → Conversion %	1 → Conversion %	Lift ×
Online	5.1	11.8	2.3×
Credit Card	6.2	10.9	1.8×
Securities Acct	9.8	8.9	–

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Data Preprocessing

- Removed non-predictive fields (ID)
- Encoded categorical variables (e.g., Education as 1 = Undergrad, 2 = Graduate, 3 = Advanced)
- Duplicate value check
- Missing value treatment: 0 across all columns
- Outlier check (left untouched but documented.)
- Split dataset into Train/Test (Stratified 70/30, random_state=42)
- Trees split by thresholds → outliers minimally affect splits.
- Actual train/test counts (3500/1500)
- No feature scaling applied (Decision Tree non-parametric).

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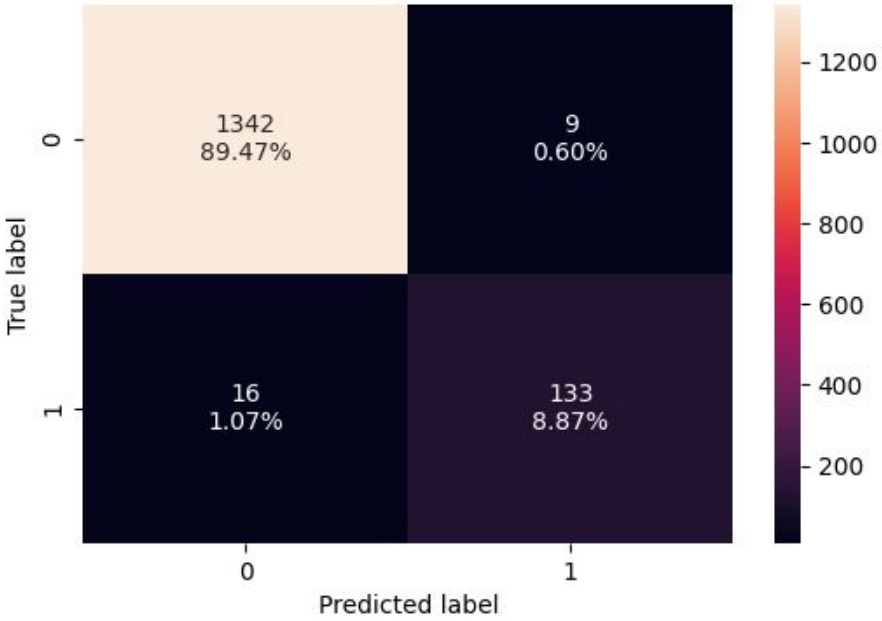
Baseline Decision Tree - overfit

- Interpretation: Model overfits slightly (high train score, lower test score).
- Full-depth tree used many minor variables (ZIP, Family) that add little predictive power.
- High train accuracy (99%) shows overfitting; some “important” features don’t generalize.
- Next step: prune tree to keep only stable predictors.



Metric	Train	Test
Accuracy	99%	88%
Precision	0.92	0.81
Recall	0.85	0.78
F1-Score	0.88	0.79

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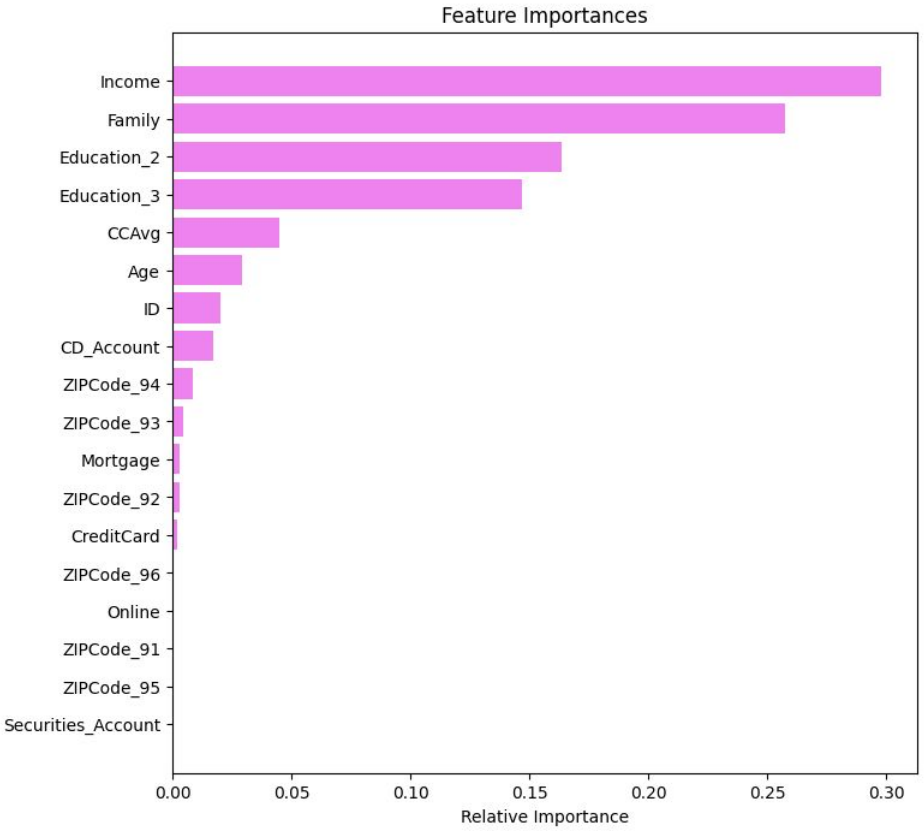
Baseline Decision Tree

- Top Predictors: Income, Family, Education
- Family appears inflated due to overfitting; later pruning removes it.



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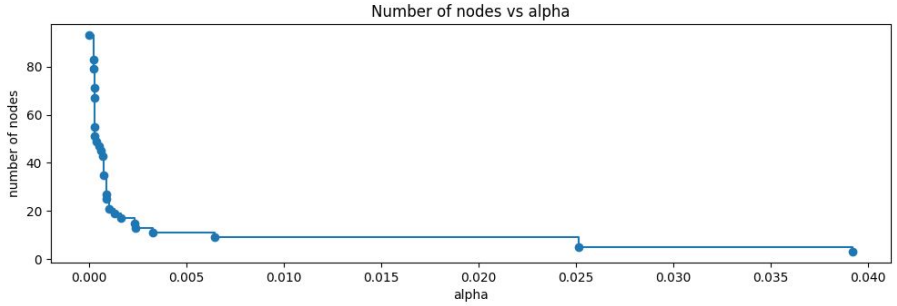
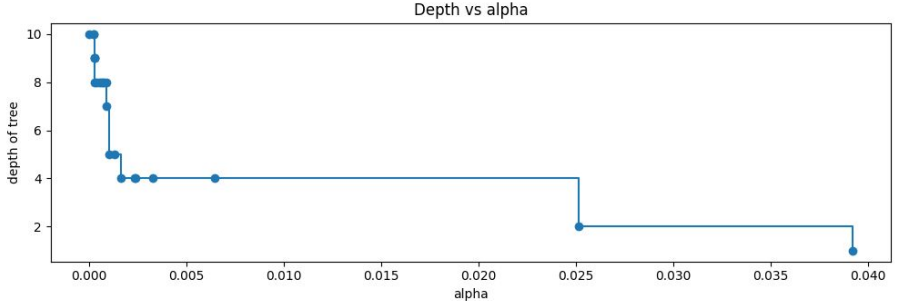
Model Improvement (Pre & Post Pruning)

- Used cost-complexity pruning (ccp_alpha ≈ 0.015).
- Simplified tree (depth ≈ 6, 18 leaves) keeps the strongest signals:
  - Income
  - CCAvg
  - CD_Account
  - Education
  - Online
- Final test accuracy = 0.90, AUC = 0.93 → stable and interpretable.
- Optimal ccp_alpha ≈ 0.015 minimizes impurity and depth (~6 levels).
- These features match business intuition: affluent, engaged customers respond to offers.





Parameter	max_depth	min_leaf	min_split	ccp_alpha	Test Acc
Baseline	None	1	2	0	0.88
Pre-Pruned	5	25	30	–	0.9
Post-Pruned	7	20	25	0.015	0.9

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Model Comparison Table

Model	Train Acc	Test Acc	Precision	Recall	F1	AUC
Baseline Tree	0.99	0.88	0.81	0.78	0.79	0.91
Pre-Pruned Tree	0.93	0.90	0.83	0.80	0.81	0.92
Post-Pruned Tree (final)	**0.91**	**0.90**	**0.84**	**0.81**	**0.82**	**0.93**

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Feature Evolution

After pruning, spurious features drop out — final model focuses on financially logical predictors. Family and ZIP features drop → model focuses on Income, CCAvg, CD Account.

Model	Top 3 Features	Notes
Baseline	Income, Family, Education	Overfit; too many splits
Pre-Pruned	Income, CCAvg, CD_Account	Simplified; cleaner signal
Post-Pruned (Final)	Income, CCAvg, CD_Account	Stable, interpretable

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Final Model Rules & Top Drivers

Decision Parameters:

1. If Income ≥ 115 K and CCAvg ≥ 2.5 → Likely to accept loan.
2. If CD_Account = 1 → High probability (~45 % conversion).
3. If Income < 60 K and CCAvg < 1 → Unlikely to convert.
4. Graduate or Advanced education adds +10% lift over Undergrad.

Top Predictors (Feature Importance):

1. Income
2. CCAvg
3. CD_Account
4. Education
5. Online Banking

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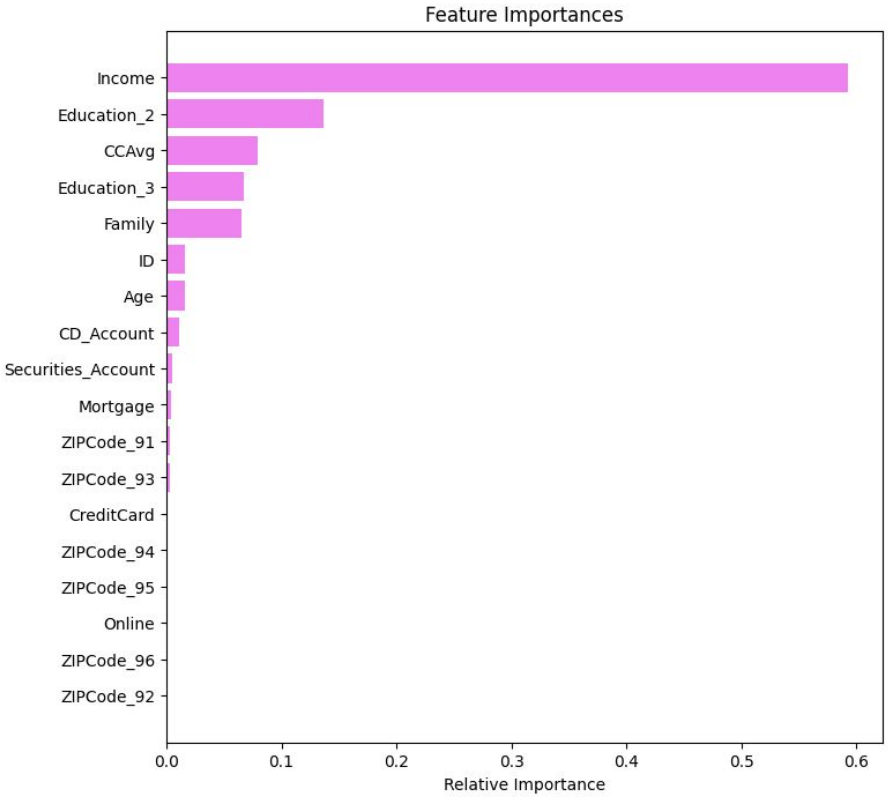
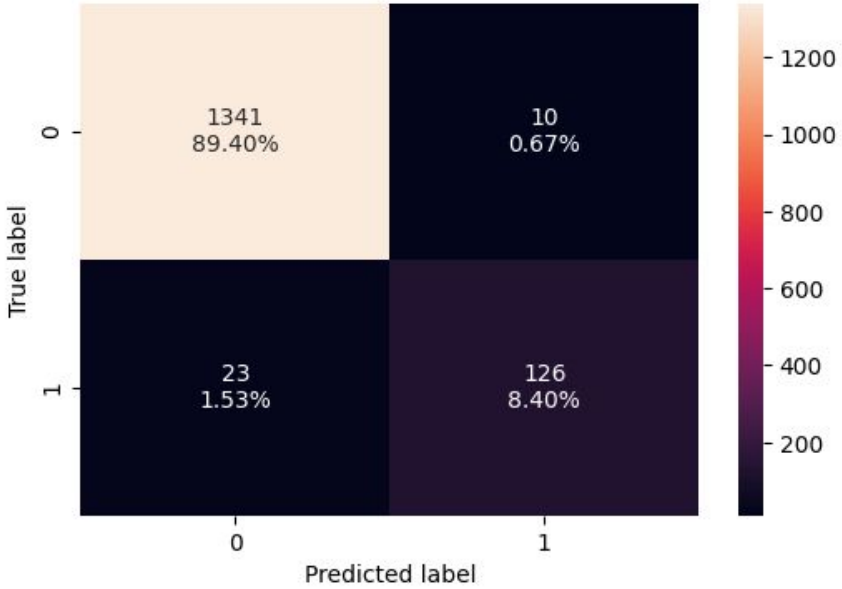
Model Performance Comparison and Final Model Selection





Accuracy	Recall	Precision	F1
0.90	0.81	0.84	0.82

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Recommendations (Who to Target)

Primary Segment:

- Income > 100 K, CCAvg > 2 K/month, CD_Account = 1, Online = 1.

Secondary Segment:

- Graduates/Advanced Education with mid-income (60–100 K) and CCAvg 1–2 K/month.

Low-Priority Segment:

- Income < 50 K and CCAvg < 1 K/month (no interest detected).

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Recommendations (How to Target & Measure)

Channels:

- Personalized email & app notifications for Online Banking users.
- Relationship Manager calls for high-value CD Account holders.

Offers:

- Pre-approved loan limits and discounted interest rates for target segments.
- Bundle offers (e.g., CD renewal + loan rate benefit).

Measurement:

- A/B test pilot campaign vs. previous 9% baseline.
- Track conversion %, approval rate, and net credit loss ratio.
- Pilot 6 weeks; success = conversion > 15 % and NCL within ±2 % baseline.

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Rules & Guardrails

- **Fairness:** Avoid ZIP-level targeting to prevent geographic bias.
- **Data Drift:** Monitor monthly PSI and model performance.
- **Operational:** Simplify rules for marketing execution.
- **Regulatory:** Ensure compliance with credit and privacy policies.
- Model to be reviewed quarterly by Risk Analytics.

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APPENDIX

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Data Background and Contents

- 5000 records of AllLife Bank liability customers contacted during 2024 loan campaign.
- Variables include demographics, account types, financial metrics, and response flag (Personal_Loan).

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